



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Vision Zero Chicago — Safety Capital Investment Prioritization

Data Analytics Capstone Project

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Period of the project

July 2026 – September 2026



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Declaration

The project titled “**Vision Zero Chicago — Safety Capital Investment Prioritization System**” has been mentored by Mr. Bhargavesh Dakka and Ms. Nithyasri Kannathal and organised by SURE Trust from July 2026 to September 2026. The project was undertaken to provide educated unemployed rural youth with hands-on experience in working on industry-relevant projects that bring them closer to prospective employers.

I declare that, to the best of my knowledge, the member(s) of the team mentioned below have worked on this project successfully and enhanced their practical knowledge in the domain of Data Analytics and Public Safety Analytics.

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Executive Summary

Project Overview

Vision Zero Chicago — Safety Capital Investment Prioritization System addresses a critical public-sector decision problem: how to allocate limited road-safety capital across Chicago's high-crash corridors to achieve the greatest safety impact while maintaining equity and fiscal discipline.

The project was developed as a transparent and reproducible decision-support system for prioritizing safety investments across 43 designated High-Crash Corridors (HCCs). Its primary objective is to support transportation planners and decision-makers in selecting the most valuable combination of safety investments under different municipal budget and equity constraints.

The analytical framework combines large-scale crash data analysis, spatial processing, statistical forecasting, engineering safety evidence, economic valuation, and mathematical optimization.

Objectives

The project was designed to:

- **Prioritize road-safety capital investments** across Chicago's 43 High-Crash Corridors.
- **Maximize life-safety benefits** under limited municipal capital budgets.
- **Support equitable investment allocation** toward socially vulnerable and historically underserved communities.
- **Provide transparent and reproducible recommendations** for selected and deferred investment opportunities.
- **Support professional decision-making** without replacing transportation engineers or government authorities.



Methodology

The project followed an end-to-end analytical methodology.

First, 877,919 traffic crash records from 2018–2025 were processed alongside Chicago street and corridor data. Spatial analysis assigned crashes to the relevant High-Crash Corridors using strict rules designed to prevent double-counting.

A 4,128 corridor-month analytical panel was then developed to estimate future crash risk. Statistical approaches including Negative Binomial regression with Empirical-Bayes shrinkage and rolling-mean forecasting models were used to forecast 2026 pre-treatment crash demand.

The project evaluated three evidence-based safety countermeasures:

1. Pedestrian Refuge Islands
2. Road Diets / 4-to-3 Lane Conversions
3. Rectangular Rapid Flashing Beacons (RRFBs)

The expected safety benefits of these treatments were estimated using FHWA Crash Modification Factors (CMFs). Crash reductions were then translated into long-term economic and societal benefits using USDOT Benefit-Cost Analysis guidance.

Finally, Mixed-Integer Linear Programming (MILP) was used to identify the optimal portfolio of safety investments under different:

- Capital budgets
- Equity spending requirements
- Safety effectiveness scenarios



Key Findings

Under the baseline planning scenario of a \$15 million budget ceiling and a 20% minimum equity spending requirement, the analysis produced the following results:

Key Performance Indicator	Result
High-Crash Corridors Funded	39 of 43
Total Capital Investment	\$14.99 Million
Budget Utilization	99.92%
Capital Directed to Equity-Priority Corridors	47.35%
Equity Investment Amount	\$7.10 Million
Fatal and Serious Injuries Avoided Annually	48.04 KSI
Total Crashes Avoided Annually	2,170.20
20-Year Societal Benefit	Approximately \$4.00 Billion
Comprehensive Societal Benefit-Cost Ratio	267.12 : 1
Direct Economic Benefit-Cost Ratio	34.2 : 1

The analysis found that treating all 43 corridors would require approximately \$17.56 million. Therefore, the \$15 million budget creates a genuine prioritization problem, resulting in four corridors being deferred under the baseline scenario.

The project also found a strong alignment between road-safety needs and social vulnerability. The optimized portfolio directed 47.35% of capital investment toward equity-priority corridors, exceeding the required 20% equity threshold.

Sensitivity analysis further identified 36 corridors as robust “Core” investments, meaning they were selected in at least 70% of the evaluated planning scenarios.



Recommendations

Based on the analytical findings, the project recommends:

1. **Advance the baseline 39-corridor portfolio** for preliminary engineering review and project scoping.
2. **Prioritize investment in high-vulnerability corridors**, where significant safety needs and equity objectives align.
3. **Conduct on-site engineering and geometric feasibility reviews** before final construction decisions are made.
4. **Use scenario-based planning** to evaluate how changes in budgets, safety assumptions, and equity requirements affect investment priorities.
5. **Maintain human oversight in all final decisions**, ensuring that analytical recommendations are validated by qualified transportation engineers and relevant government authorities.

Executive and Stakeholder Highlights

Executive Dimension	Key Takeaway
Capital Efficiency	\$14.99M of the \$15M budget was allocated, achieving 99.92% budget utilization.
Life-Safety Impact	The baseline portfolio is estimated to prevent 48.04 fatal and serious injury crashes annually.
Equity	47.35% of capital is directed toward equity-priority corridors.
Economic Value	The portfolio produces an estimated \$4.00B in long-term societal benefits.
Investment Robustness	36 corridors remain high-priority investments across multiple planning scenarios.



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Executive Dimension	Key Takeaway
Transparency	Recommendations are based on documented data, assumptions, safety evidence, and optimization constraints.
Governance	The system supports decision-making; final engineering and investment authority remains with qualified professionals.

Conclusion

Vision Zero Chicago — Safety Capital Investment Prioritization System demonstrates how Data Analytics, spatial analysis, statistical forecasting, benefit-cost analysis, and mathematical optimization can be integrated to address a complex public-sector investment problem.

The project provides a structured and evidence-based approach for prioritizing limited safety capital across Chicago's High-Crash Corridors. Rather than relying solely on subjective prioritization, the system enables stakeholders to evaluate investment decisions based on expected safety impact, budget constraints, economic value, and equity considerations.

Most importantly, the system is designed as a decision-support framework rather than an automated decision-maker. Final infrastructure decisions remain subject to professional engineering review, physical feasibility assessment, and public-sector governance.



Introduction

Traffic safety is a major urban planning and public policy challenge. Between 2018 and 2025, the City of Chicago recorded more than 877,000 traffic crashes, resulting in fatalities, serious injuries, and substantial social and economic costs. These crashes are not evenly distributed across the city. Severe crashes tend to concentrate along particular arterial corridors where factors such as high traffic volumes, vehicle speeds, multiple travel lanes, and challenging pedestrian conditions can increase road-safety risks.

In response to this challenge, the City of Chicago established the Vision Zero initiative, based on the principle that traffic fatalities and serious injuries should be prevented through safer transportation systems and evidence-based interventions. As part of this broader road-safety context, 43 High-Crash Corridors (HCCs) were identified as priority areas requiring focused safety attention.

The central problem addressed by this project is the efficient allocation of limited safety capital. Municipal agencies cannot fund improvements across every high-crash corridor at the same time. Decision-makers must therefore determine which investments should receive priority while considering expected safety benefits, available budgets, economic value, and equitable distribution of resources.

Vision Zero Chicago — Safety Capital Investment Prioritization System was developed to address this decision problem through a structured and transparent analytical framework. The governing question of the project is:

Which combination of corridor-level road-safety investments should be prioritized for engineering review under limited capital budgets while meeting defined equity requirements?

The project aims to analyze historical crash patterns across Chicago's 43 High-Crash Corridors, estimate baseline crash risk, evaluate evidence-based safety interventions, quantify their potential benefits, and identify an optimized portfolio of investments. The system uses spatial data analysis, statistical forecasting, safety treatment evaluation, benefit-cost analysis, and Mixed-Integer Linear Programming (MILP) to support this process.

The analytical scope of the project covers 43 High-Crash Corridors in Chicago and uses historical crash data from 2018 to 2025, representing 96 monthly periods, to estimate pre-treatment crash risk for 2026. The project evaluates three infrastructure-based safety countermeasures: Pedestrian Refuge



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Islands, Road Diets or 4-to-3 Lane Conversions, and Rectangular Rapid Flashing Beacons (RRFBs). Their potential safety effects are evaluated using established FHWA Crash Modification Factors, while the long-term economic and societal value of crash reduction is estimated using a 20-year analysis period and USDOT Benefit-Cost Analysis guidance.

The project operates within clearly defined limitations and governance boundaries. It is a decision-support system and not an automated project approval system. Final decisions regarding infrastructure investment, engineering design, and construction remain the responsibility of qualified transportation engineers and relevant government authorities. Physical feasibility considerations, including roadway geometry, utilities, transit operations, and other site-specific constraints, require further engineering and field verification before implementation.

The analysis also has data and methodological limitations. Continuous traffic exposure measures such as corridor-level AADT are not available across the complete analytical network, meaning the project forecasts recorded crash counts rather than exposure-adjusted crash rates. In addition, the estimated treatment benefits are based on FHWA Crash Modification Factors derived from broader empirical evidence and should not be interpreted as locally observed causal results from randomized experiments.

An important innovation of the project is the integration of multiple analytical disciplines into a single decision-support framework. Rather than treating crash analysis, forecasting, economic evaluation, equity, and capital planning as separate activities, the project connects them through an end-to-end analytical pipeline.

Another key innovation is the treatment of equity as a measurable planning constraint rather than only a reporting metric. Equity spending requirements based on the Social Vulnerability Index (SVI) are incorporated directly into the investment optimization process, ensuring that selected portfolios can be evaluated against defined equity objectives.

The project also supports scenario-based planning by allowing decision-makers to examine how changes in available budgets, equity requirements, and safety assumptions affect investment priorities. This makes the system more flexible than a traditional static report and enables stakeholders to understand the trade-offs associated with different planning decisions.



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Finally, transparency and governance are central components of the project. The system provides documented analytical logic and distinguishes between recommendations generated by the model and decisions requiring professional judgment. By combining data analytics, transportation safety evidence, economic evaluation, and mathematical optimization within a transparent governance framework, Vision Zero Chicago — Safety Capital Investment Prioritization System demonstrates how modern analytics can support more objective, evidence-based, and accountable public-sector capital planning.



Project Objectives

Section	Required Content	Vision Zero Chicago Content
Executive Summary	Concise overview of objectives, methods, findings, and recommendations	Explains the capital-allocation challenge, the DSS solution, baseline portfolio results, key analytical distinctions, and governance boundaries.
Introduction	Background and context	Establishes Chicago's traffic-safety challenge, Vision Zero context, 43 High-Crash Corridors, and the relationship between crash risk and equity.
Introduction	Problem statement and goals	Defines the core decision: which corridor investments should be prioritized under limited budgets and equity requirements?
Introduction	Scope and limitations	Covers the 43 corridors, 2018–2025 crash data, three treatments, forecasting horizon, economic valuation, and engineering/governance limitations.
Introduction	Innovation component	Highlights the integrated GIS-to-optimization pipeline, mathematical equity constraints, interactive what-if planning, and governance architecture.
Project Objectives	Clearly defined objectives	Covers spatial data quality, crash-risk forecasting, treatment valuation, portfolio optimization, and transparent decision support.
Project Objectives	Expected outcomes	Focuses on defensible capital allocation, equitable investment distribution, and improved life-safety outcomes.
Project Objectives	Deliverables	Includes the data pipeline, predictive models, treatment/BCA matrix, optimization engine, Streamlit dashboard, and governance/verification framework.



Methodology and Results

The Vision Zero Chicago Decision Support System was developed as an end-to-end analytical framework that transforms historical traffic crash data into transparent and evidence-based capital investment recommendations. The methodology integrates geospatial analysis, statistical forecasting, transportation safety evaluation, economic benefit-cost analysis, and mathematical optimization. The final system supports decision-makers in determining which road-safety investments should be prioritized across Chicago's 43 designated High-Crash Corridors under different budget and equity constraints.

Methods and Technology Used

The project applies a structured analytical methodology consisting of five major stages: data collection and preparation, geospatial processing, crash-risk forecasting, countermeasure valuation, and portfolio optimization.

Geospatial Analysis and Crash Assignment

All geographic datasets, including crash locations, roadway centerlines, and corridor geometries, were standardized using EPSG:3435, NAD83 Illinois State Plane East, a local coordinate system suitable for accurate distance-based spatial analysis.

A deterministic nearest-centerline methodology was used to assign crashes to the appropriate High-Crash Corridor. A 100-foot corridor buffer was applied during the screening process, and crashes associated with multiple candidate corridors were assigned to only one nearest centerline. This approach was designed to prevent duplicate crash assignments and maintain a clear audit trail.

Time-Series Data Preparation and Forecasting

Historical crash data from **2018 to 2025** was transformed into a balanced corridor-month analytical panel covering:

- 43 High-Crash Corridors
- 96 monthly periods
- 4,128 corridor-month observations

The forecasting methodology followed a chronological validation structure to reduce the risk of data leakage:



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Period	Purpose
2018	Warm-up period for historical lag calculations
2019–2023	Model training
2024	Validation and model selection
2025	Out-of-time holdout testing
2026	Forecast period

For total crashes, multiple approaches were evaluated, including statistical models and a trailing rolling-mean benchmark. For fatal and serious injury crashes, a Negative Binomial Generalized Linear Model combined with Empirical-Bayes shrinkage was used to provide more stable estimates for low-frequency but high-impact crash events.

Safety Countermeasure Evaluation

The system evaluated three road-safety infrastructure treatments using Federal Highway Administration Crash Modification Factors (CMFs):

Treatment	Crash Modification Factor	Estimated Crash Reduction
Pedestrian Refuge Islands	0.68	32%
Road Diet / 4-to-3 Lane Conversion	0.71	29%
Rectangular Rapid Flashing Beacons (RRFBs)	0.53	47%

The estimated crash reductions were applied to the forecasted pre-treatment crash burden for each corridor. This allowed the project to estimate the potential number of crashes and severe injuries that could be prevented through different infrastructure investments.

Economic Benefit-Cost Analysis

The estimated safety benefits were translated into economic value using USDOT crash-cost guidance across KABCO injury severity categories.

The analysis used:

- A 20-year evaluation period
- A 3% real annual discount rate



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- Comprehensive societal crash-cost estimates

The resulting calculations estimate the societal value of fatalities, injuries, emergency response, medical impacts, property damage, and other consequences avoided through safety improvements.

It is important to distinguish these estimates from municipal revenue. The calculated benefits represent societal economic harms avoided, rather than cash income or direct financial returns to the City of Chicago.

Mathematical Portfolio Optimization

The final investment portfolio was selected using Mixed-Integer Linear Programming (MILP) with the HiGHS optimization solver.

The optimization model seeks to maximize expected societal safety benefits while satisfying real-world planning constraints.

The major constraints include:

- Available capital budget
- Mandatory equity spending requirements
- Maximum of one selected treatment per corridor
- Economic viability requirements
- Portfolio diversification rules

The system evaluated planning scenarios using budget ceilings of:

- **\$15 million**
- **\$25 million**
- **\$40 million**

Equity spending floors of:

- **20%**
- **30%**
- **40%**

were also incorporated into the optimization framework for high-SVI priority areas.

Tools and Software Used

The project was developed primarily in Python and uses specialized analytical, geospatial, optimization, visualization, and deployment technologies.



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Category	Tool / Software	Purpose
Programming Language	Python	Core analytical and development environment
Data Analysis	Pandas and NumPy	Data cleaning, transformation, and aggregation
Statistical Modeling	Statsmodels	Negative Binomial regression modeling
Geospatial Analysis	GeoPandas and Shapely	Spatial processing and crash assignment
Optimization	SciPy / HiGHS	Mixed-Integer Linear Programming
Database Processing	DuckDB and SQLite	Analytical data storage and querying
Dashboard Development	Streamlit	Interactive decision-support application
Data Visualization	Plotly	Interactive charts and analytical visuals
GIS Visualization	PyDeck / Mapbox GL	Interactive corridor mapping
Testing	Pytest	Automated validation and quality testing
Version Control	GitHub	Source-code and project management

Data Collection Approach

The project uses publicly available and authoritative datasets from municipal and federal sources.

The primary data collection process included traffic crash records, roadway spatial data, social vulnerability information, and transportation safety evidence.

Source	Data Collected	Project Use
City of Chicago Data Portal	Traffic crash records	Crash analysis and forecasting
City of Chicago Data Portal	Vehicle and people datasets	Injury and severity analysis
Chicago Street Centerlines	Road network geometry	Spatial corridor analysis
CDOT Vision Zero Framework	High-Crash Corridor locations	Project geography
CDC/ATSDR SVI	Social vulnerability indicators	Equity prioritization
FHWA CMF Clearinghouse	Treatment effectiveness estimates	Safety benefit modeling
USDOT BCA Guidance	Crash economic values	Benefit-cost analysis



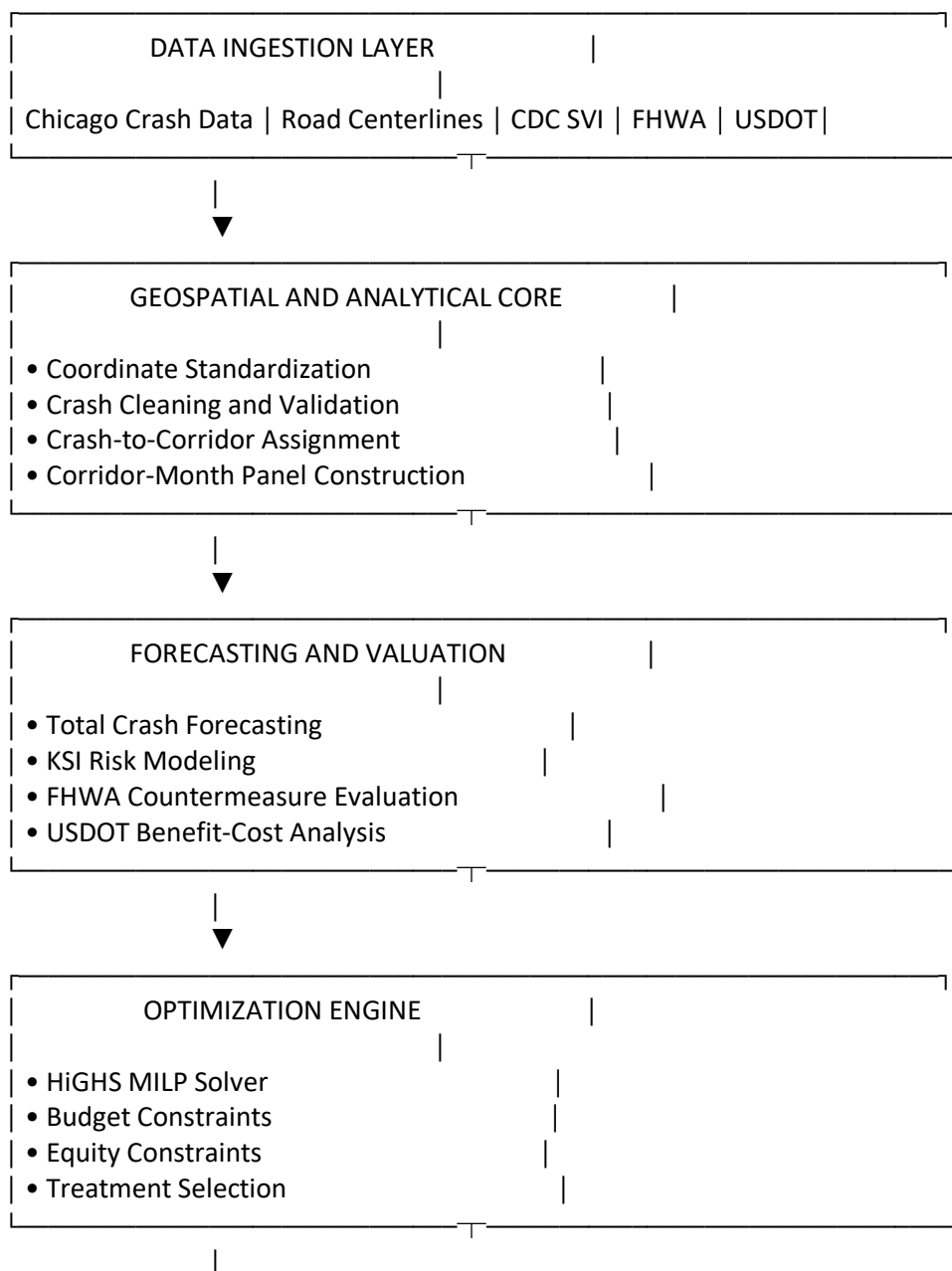
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The analytical dataset includes more than 877,000 historical crash records from 2018 to 2025.

The data was processed through structured quality checks before entering the analytical pipeline. Spatially invalid or ambiguous records were handled through explicit validation and audit processes to preserve transparency and reproducibility.

Project Architecture

The Vision Zero Chicago system follows a five-tier architecture that separates raw data processing from analytical modeling, optimization, and stakeholder delivery.





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DECISION-SUPPORT APPLICATION	
• Executive Dashboard	
• Portfolio Analysis	
• What-If Planner	
• Corridor GIS Explorer	
• Governance and Audit Information	

The architecture creates a clear analytical flow:

Raw Data → Data Validation → Spatial Analysis → Risk Forecasting → Safety Valuation → Portfolio Optimization → Decision Dashboard

This separation improves reproducibility and makes the system easier for stakeholders and reviewers to understand and audit.

Results and Final Project Output

The baseline scenario evaluated a **\$15 million capital budget** with a **20% equity spending floor**.

The optimization model produced the following results:

Key Performance Indicator	Baseline Result
Budget Available	\$15.0 Million
Capital Invested	\$14.99 Million
Budget Utilization	99.92%
Remaining Budget Slack	Approximately \$11,490
Corridors Funded	39 of 43
Corridors Deferred	4
Capital Directed to High-SVI Areas	47.35%
KSI Avoided per Year	48.04



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Key Performance Indicator	Baseline Result
Total Crashes Avoided per Year	2,170.20
20-Year Societal Benefit	Approximately \$4.00 Billion
Societal Benefit-Cost Ratio	267.12 : 1
Direct Economic Benefit-Cost Ratio	34.2 : 1

The results demonstrate that the **\$15 million budget is a binding constraint**. Treating all 43 corridors would require approximately **\$17.56 million**, requiring the optimization model to prioritize investments. Under higher budget scenarios of **\$25 million and \$40 million**, the system can fund the full network of 43 corridors.

Final Project Working Screens

The completed Vision Zero Chicago application contains four primary stakeholder interfaces.

Executive Recommendation

The Executive Recommendation page provides senior stakeholders with the most important portfolio metrics.

It presents:

- Total capital investment
- Number of corridors funded
- Estimated KSI reduction
- Equity capital allocation
- Budget utilization
- Overall societal benefit

The baseline portfolio demonstrates that **39 corridors can be prioritized within the available \$15 million budget while exceeding the required equity threshold**.



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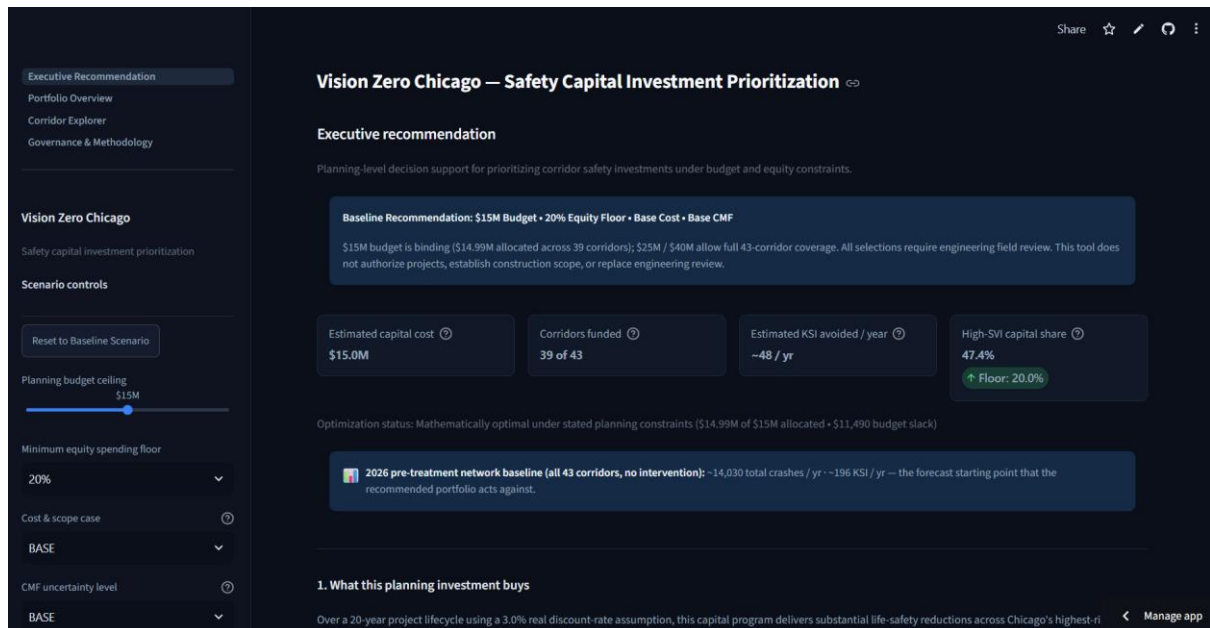


Figure 1: Executive dashboard showing the recommended baseline investment portfolio and key safety performance indicators.

Portfolio Overview and What-If Planner

The Portfolio Overview allows users to understand portfolio performance and evaluate alternative scenarios.

Users can modify:

- Capital budget
- Equity requirements
- Planning assumptions

The system then recalculates the optimal investment portfolio.

This feature allows stakeholders to understand the trade-offs between available funding, equity policy, and expected safety outcomes.



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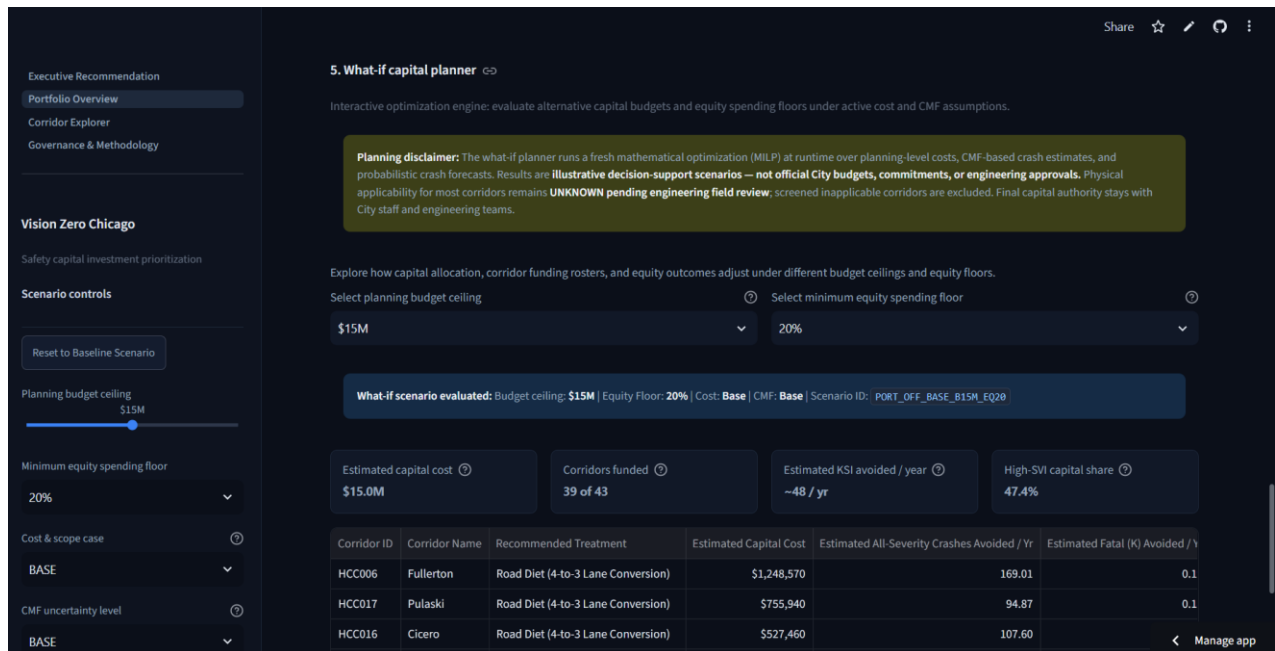


Figure 2: Interactive portfolio analysis and capital planning interface.

Corridor Explorer and GIS Map

The Corridor Explorer provides detailed information for each of the 43 High-Crash Corridors.

Users can review:

- Crash-risk forecasts
- Recommended treatment
- Estimated project cost
- Expected KSI reduction
- Benefit-Cost Ratio
- Equity classification
- Investment selection rationale



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The GIS map provides a spatial representation of funded and deferred corridors across Chicago.

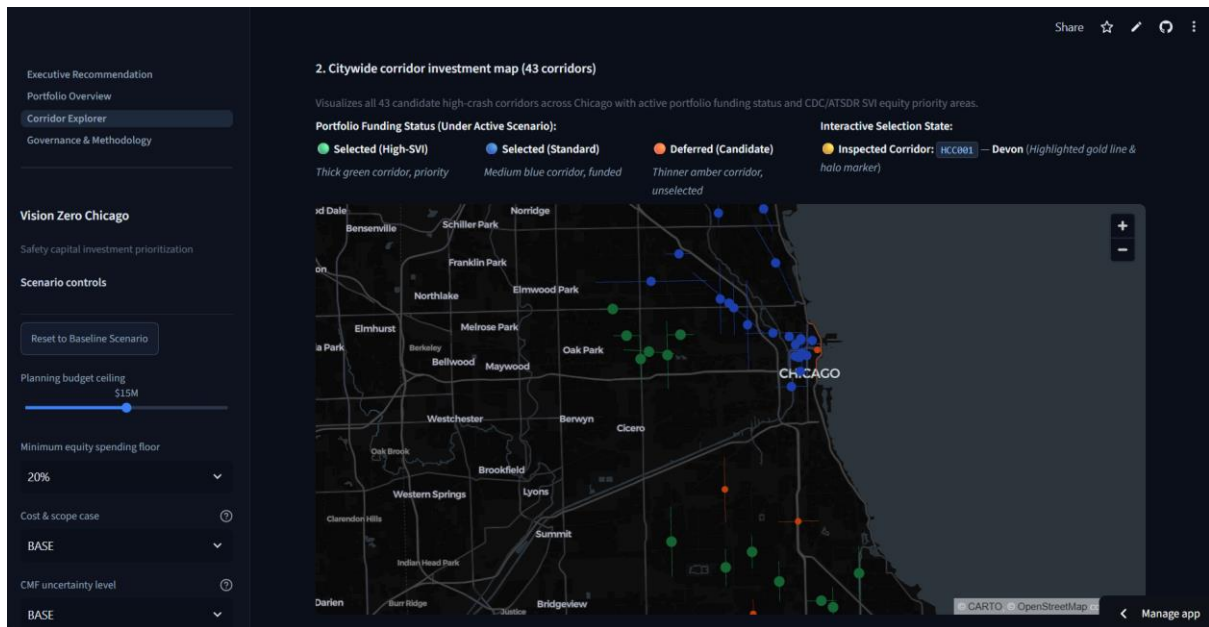


Figure 3: Interactive corridor-level analysis and spatial investment map.

Governance and Methodology

The Governance and Methodology interface provides transparency regarding the analytical process and decision boundaries.

It documents:

- Data and pipeline lineage
- Analytical assumptions
- Validation information
- Warning and limitation registers
- Governance boundaries

The system explicitly recognizes that it is a decision-support tool.

Final engineering feasibility and capital approval remain the responsibility of City officials and professional transportation engineers.



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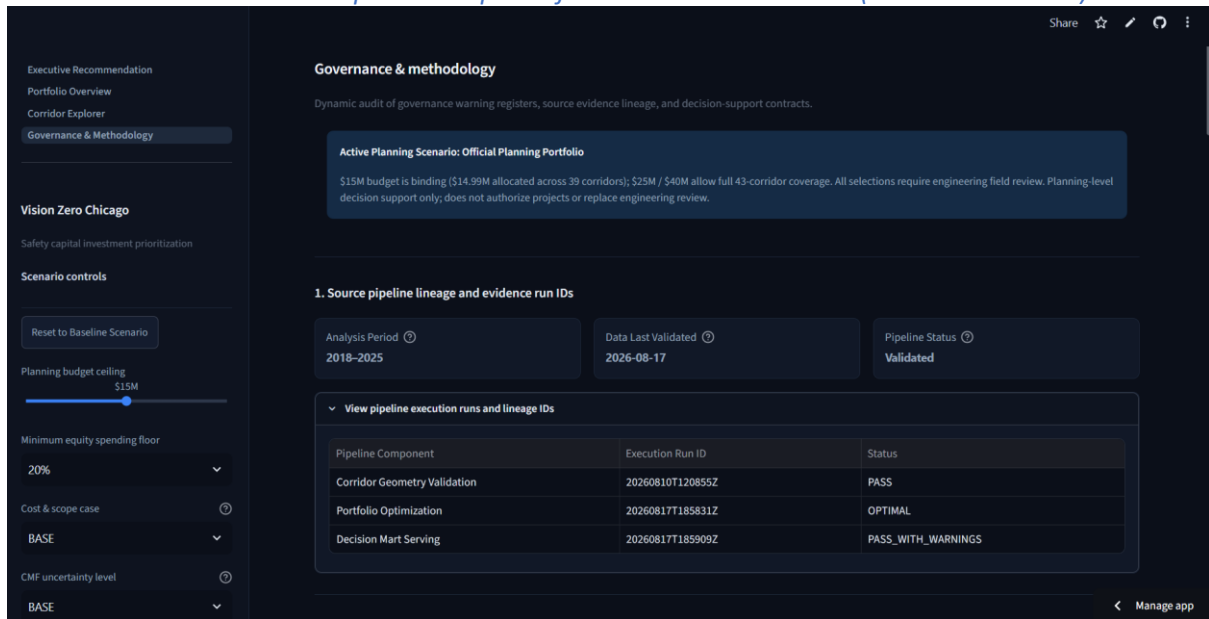


Figure 4: Governance, methodology, and analytical lineage interface.

Project GitHub Link

The complete source code, project structure, analytical pipeline, and supporting documentation are available through the project's GitHub repository:

Vision Zero Chicago GitHub Repository : <https://github.com/CharanKumar17-a/vision-zero-chicago>

Live Project Application

The deployed interactive Decision Support System is available at:

<https://vision-zero-chicago-charan.streamlit.app/>

Conclusion

The Vision Zero Chicago project demonstrates how multiple analytical disciplines can be integrated into a single decision-support system for public-sector capital planning.

The complete methodology connects:

Historical Crash Data → Geospatial Analysis → Risk Forecasting → Countermeasure Evaluation → Economic Valuation → Mathematical Optimization → Interactive Decision Support.



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The final system provides a transparent and reproducible framework for prioritizing limited road-safety investments. Rather than replacing professional engineering judgment, the system provides decision-makers with evidence-based recommendations, explicit trade-offs, and corridor-level rationale to support more effective and equitable transportation safety planning.



Social / Industry Relevance Of The Project

The Vision Zero Chicago Decision Support System demonstrates how modern data analytics, geospatial technology, statistical forecasting, and mathematical optimization can be applied to a real-world public-sector transportation safety challenge. The project is relevant both socially and professionally because it addresses the allocation of limited road-safety investment toward locations where interventions can potentially deliver the greatest life-safety benefits while considering equity requirements.

Social Relevance

Road traffic crashes have significant consequences for individuals, families, and communities. Fatalities and serious injuries can result in long-term physical, emotional, and economic impacts. The Vision Zero Chicago project addresses this public safety challenge by developing an evidence-based framework for prioritizing safety investments across Chicago's high-crash corridors.

Rather than relying solely on reactive responses to individual crashes or subjective prioritization, the system combines historical crash data, risk forecasting, safety countermeasure evaluation, and portfolio optimization. This provides decision-makers with a structured approach for identifying investment opportunities under limited capital budgets.

A major social contribution of the project is the integration of equity into the investment decision framework. The system evaluates capital allocation across areas with higher social vulnerability and applies equity spending requirements during portfolio optimization. This helps ensure that vulnerable and historically underserved communities are explicitly considered when prioritizing road-safety investments.

The project therefore supports the broader Vision Zero principle that traffic deaths and serious injuries should be addressed through systematic, evidence-based improvements to transportation systems.

Industry Relevance

The project has strong relevance across multiple professional and technical fields, including:

- Data Analytics and Business Analytics
- Data Science and Statistical Modeling



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- Transportation and Urban Planning
- Civil and Transportation Engineering
- Geographic Information Systems (GIS)
- Operations Research
- Public-Sector Decision Support Systems.

The Vision Zero Chicago system demonstrates how these disciplines can work together within a single analytical framework. Large-scale municipal crash data is processed and prepared using data engineering practices, spatial relationships are analyzed using GIS, future crash risk is estimated using statistical forecasting models, safety interventions are evaluated using documented Crash Modification Factors, and investment portfolios are prioritized using Mixed-Integer Linear Programming.

This multidisciplinary approach reflects the increasing importance of integrated analytics in solving complex industry and public-sector problems.

Relevance to Data-Driven Decision-Making

Organizations frequently face situations where resources are limited and multiple competing projects must be evaluated. Traditional approaches may depend heavily on spreadsheets, subjective judgment, or isolated performance indicators.

The Vision Zero Chicago Decision Support System demonstrates a more integrated approach by connecting:

Historical Data → Risk Forecasting → Safety Intervention Evaluation → Economic Valuation → Portfolio Optimization → Decision Support

This framework enables decision-makers to understand not only which projects are recommended, but also the analytical rationale behind those recommendations.

The system provides transparent information about:

- Corridor-level crash risk



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- Estimated safety benefits
- Capital costs
- Budget constraints
- Equity allocation
- Reasons for project selection or deferral

This makes the project particularly relevant to organizations seeking transparent, reproducible, and evidence-based approaches to capital allocation.

Public-Sector and Transportation Industry Relevance

Public-sector organizations must often justify how limited taxpayer and grant-funded resources are allocated. Transportation agencies, city governments, and infrastructure planners require analytical methods that can balance safety outcomes, financial limitations, policy requirements, and engineering constraints.

The project demonstrates how Mixed-Integer Linear Programming (MILP) can support this type of capital planning problem. Instead of simply ranking corridors by crash counts or benefit-cost ratios, the optimization framework evaluates combinations of investments while considering multiple constraints simultaneously.

The system is also designed with an important governance boundary:

The system supports decisions; it does not approve projects.

Final engineering feasibility, project design, and construction authorization remain the responsibility of qualified transportation engineers and relevant City authorities. This ensures that analytical recommendations complement rather than replace professional expertise.

Broader Contribution of the Project

The broader contribution of the Vision Zero Chicago project is the demonstration of an end-to-end analytical architecture that can be adapted to other infrastructure and public-sector decision problems.

The approach integrates:



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Municipal Open Data → Geospatial Analytics → Predictive Modeling → Safety Valuation →

Mathematical Optimization → Interactive Decision Support

Similar analytical frameworks could potentially be applied to other cities and sectors where decision-makers must allocate limited resources across competing investment opportunities.

By combining technical rigor with transparency and human governance, the Vision Zero Chicago Decision Support System demonstrates the practical value of analytics beyond reporting and visualization. It shows how data-driven methods can support complex, high-impact decisions while maintaining accountability, reproducibility, and professional oversight.

Summary

The project has significant social relevance because it focuses on improving road safety and supporting equitable investment in vulnerable communities. It has strong industry relevance because it integrates data analytics, GIS, statistical forecasting, Benefit-Cost Analysis, and Operations Research into a single decision-support framework.

Most importantly, the project demonstrates how modern analytics can help organizations make more transparent, evidence-based, and defensible investment decisions without replacing human judgment or professional authority.



The Vision Zero Chicago Decision Support System provided significant learning across data analytics, geospatial analysis, predictive modeling, optimization, software development, and project management. The project also provided practical experience in transforming a complex real-world public policy problem into a structured and transparent analytical decision-support system.

Technology and Technical Learning

One of the major learning outcomes was understanding how multiple technologies can be integrated into a single end-to-end analytics system.

The project provided practical experience in:

- **Python-based data analytics and engineering** for cleaning, transforming, and validating large-scale crash datasets.
- **Geospatial analytics using GeoPandas and Shapely**, including spatial joins, corridor matching, coordinate reference systems, and nearest-centerline assignment.
- **EPSG:3435 spatial processing**, which enabled distance-based analysis using an appropriate local coordinate system.
- **Time-aware predictive modeling**, particularly the importance of chronological validation and preventing future data leakage.
- **Empirical-Bayes and statistical modeling approaches** for estimating rare but high-impact crash outcomes such as fatal and serious injury crashes.
- **Mixed-Integer Linear Programming (MILP)** and the use of the **HiGHS solver** to optimize investment portfolios under budget and equity constraints.
- **Streamlit application development** to transform analytical outputs into an interactive decision-support tool.
- **Interactive GIS visualization**, enabling stakeholders to explore corridor-level results spatially.
- **Automated testing and data-quality validation**, improving reproducibility and reducing the risk of errors across the analytical pipeline.



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A particularly important technical learning was that a more complex model is not automatically a better model. The project emphasized the importance of comparing advanced approaches against simple baseline methods and selecting models based on honest validation performance rather than algorithmic complexity.

Analytical and Problem-Solving Learning

The project strengthened the ability to move beyond descriptive analytics toward decision-focused analytics.

Instead of simply answering:

Which corridors have the highest number of crashes?

the project addressed the more practical decision:

Which combination of safety investments should be prioritized under a limited budget while meeting equity requirements?

This required combining several analytical disciplines into one decision framework:

Historical Data → Risk Forecasting → Countermeasure Evaluation → Economic Valuation → Portfolio Optimization → Decision Support

The project demonstrated that ranking projects individually is not sufficient when decisions involve limited budgets and multiple constraints. A corridor with the highest individual benefit may not necessarily belong to the optimal overall portfolio.

Management and Project Development Learning

The project also provided experience in managing a complex analytics workflow with multiple dependencies.

Important management learnings included:

- Breaking a large project into structured stages.
- Defining clear analytical objectives before building models.
- Validating data before producing dashboards or recommendations.
- Maintaining reproducible workflows.



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- Documenting assumptions and methodological limitations.
- Separating verified results from design assumptions.
- Using automated quality checks to prevent regression.
- Maintaining governance boundaries between analytical recommendations and professional engineering decisions.

One of the strongest lessons was the importance of decision governance. The analytical system can identify and prioritize potential investments, but it should not claim authority to approve physical infrastructure projects. Engineering feasibility, field conditions, utilities, roadway geometry, and final project authorization remain human and professional responsibilities.

Overall Project Experience

Working on Vision Zero Chicago provided practical experience with the realities of large-scale public data.

Real-world crash data is substantially more complex than typical academic datasets. The project involved working with a large volume of historical records and addressing issues related to data quality, spatial matching, inconsistent records, and analytical uncertainty.

The experience demonstrated the importance of not silently removing problematic records. Instead, data-quality issues need to be identified, documented, and handled through transparent validation processes wherever possible.

Another important experience was balancing efficiency and equity.

A purely financial or benefit-maximization approach may not fully represent public policy priorities. The project therefore explored how equity requirements could be incorporated directly into the portfolio optimization framework rather than being treated only as a reporting metric after the investment decisions were made.

This provided practical experience in translating policy requirements into mathematical constraints.

Key Reflection

The most important learning from the project was that successful analytics is not only about building a technically sophisticated model.

A useful real-world analytics system must be:



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- **Accurate enough for the decision**
- **Transparent**
- **Reproducible**
- **Validated**
- **Explainable to stakeholders**
- **Aligned with operational constraints**
- **Clear about its limitations**

The Vision Zero Chicago project demonstrated how data analytics, geospatial technology, statistical forecasting, optimization, and interactive software can work together to support complex public-sector decisions.

The overall experience strengthened practical skills in both technical analytics and decision-making, while reinforcing the importance of responsible AI and analytics governance.

Final Reflection: The value of an analytical decision-support system is not in replacing human decision-makers. Its value is in providing transparent, evidence-based, and reproducible information that helps decision-makers make better and more defensible decisions.

Individual Contribution and Learning Summary

Learning Area	Experience and Learning
Data Analytics	Learned to process, clean, validate, and analyze large-scale real-world datasets.
Geospatial Analysis	Developed practical understanding of coordinate systems, spatial matching, corridor analysis, and GIS visualization.
Predictive Modeling	Learned the importance of baseline models, chronological validation, and preventing data leakage.
Optimization	Applied MILP to solve a real-world capital allocation problem under multiple constraints.



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Learning Area	Experience and Learning
Software Development	Developed experience in building an interactive Streamlit-based decision-support application.
Data Quality	Learned the importance of validation, automated testing, auditability, and reproducibility.
Project Management	Developed a structured, dependency-based approach to managing a complex analytics project.
Business and Policy Thinking	Learned to translate a public-sector policy problem into measurable objectives and mathematical decision constraints.
Stakeholder Communication	Learned to present complex analytical results through executive summaries, dashboards, visualizations, and transparent rationale.
Professional Growth	Developed a stronger understanding that analytics must balance technical rigor, practical feasibility, governance, and real-world decision value.



Conclusion and Future Scope

The Vision Zero Chicago Decision Support System was developed to address a practical municipal decision: how to prioritize limited road-safety capital across Chicago's high-crash corridors while balancing life-safety benefits, budget constraints, and equity requirements.

The project successfully transformed this complex problem into an end-to-end analytical decision-support framework. Historical crash data was processed and spatially assigned to the project's 43 high-crash corridors, baseline crash risk was forecast, safety countermeasures were evaluated using documented Crash Modification Factors, and investment portfolios were optimized using mathematical programming.

Recap of Objectives and Achievements

The project achieved its primary objectives across five areas:

Project Objective	Achievement
Data and Spatial Analysis	Processed more than 877,000 historical crash records and developed a structured corridor-level analytical dataset for the 43 High-Crash Corridors.
Risk Forecasting	Developed time-aware forecasting methods to estimate 2026 pre-treatment crash and fatal/serious injury risk.
Countermeasure Evaluation	Evaluated Pedestrian Refuge Islands, Road Diets, and RRFBs using documented FHWA Crash Modification Factors and economic valuation methods.
Portfolio Optimization	Applied Mixed-Integer Linear Programming using the HiGHS solver to identify investment portfolios under different budget and equity constraints.
Decision-Support Deployment	Developed and deployed an interactive Streamlit application for stakeholders to explore portfolio results, corridor-level decisions, and what-if scenarios.

Under the baseline planning scenario of a \$15 million budget ceiling and a 20% equity spending floor, the system identified a portfolio that:



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- Funds 39 of 43 high-crash corridors.
- Allocates approximately \$14.99 million of the available capital budget.
- Directs approximately 47.35% of investment toward high-SVI equity-priority corridors.
- Estimates approximately 48 fatal and serious injuries avoided annually.
- Estimates approximately 2,170 total crashes avoided annually.
- Produces an estimated \$4.00 billion in 20-year societal benefits, based on the project's documented benefit-cost methodology.

These results demonstrate the main value of the system: it provides a structured and reproducible method for comparing investment alternatives rather than relying solely on isolated rankings, subjective prioritization, or reactive decision-making.

The completed project also resulted in a live interactive application:

Live Vision Zero Chicago Decision Support System:

<https://vision-zero-chicago-charan.streamlit.app/>

Future Scope of the Project

Although the current system provides a complete decision-support framework, several opportunities exist for future development.

1. Integration of Traffic Exposure Data

A major future enhancement would be the integration of continuous traffic volume data, such as **Average Annual Daily Traffic (AADT)**.

This would allow the analysis to move beyond aggregate crash counts and incorporate exposure-based safety measures, such as crash rates relative to vehicle activity or vehicle-miles traveled.

Potential improvement:

Crash Data + Traffic Exposure + Road Characteristics → More refined risk estimation

2. Expansion Beyond the Current High-Crash Corridors

The current project focuses on Chicago's **43 designated High-Crash Corridors**.



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Future versions could expand the system to analyze a larger portion of Chicago's transportation network and automatically identify emerging high-risk locations. This could include:

- School zones
- Transit corridors
- Bicycle corridors
- Pedestrian-heavy intersections
- Emerging crash hotspots

This would allow the system to evolve from prioritizing a predefined network into a more dynamic citywide safety-monitoring framework.

3. Automated Engineering Feasibility Assessment

The current system provides decision support but does not replace engineering field review.

A future version could integrate additional infrastructure and asset data, including:

- Lane widths
- Roadway geometry
- Median availability
- Transit routes
- Intersection layouts
- Utility constraints
- Existing pedestrian infrastructure

This could provide an earlier automated screening of whether particular countermeasures are potentially feasible before detailed field investigation.

4. Expanded Safety Countermeasure Library

The current analysis evaluates three infrastructure treatments. Future development could expand the candidate treatment library to include additional safety interventions, such as:

- Protected bicycle lanes



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- Curb extensions
- Leading Pedestrian Intervals
- Signal improvements
- Speed management infrastructure
- Pedestrian safety improvements
- Intersection redesigns

An expanded treatment library would provide planners with a broader range of investment alternatives.

5. Multi-Year Capital Investment Planning

The current optimization framework primarily evaluates portfolio decisions within planning budget scenarios.

A future version could extend the optimization model into a **multi-year capital programming system**.

For example:

Year 1 → Highest-priority projects

Year 2 → Remaining high-benefit projects

Year 3 → Network expansion projects

The optimization could consider:

- Annual budget availability
- Construction schedules
- Project dependencies
- Inflation
- Construction capacity
- Geographic coordination between nearby projects

This would make the system more directly applicable to long-term capital improvement planning.



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6. Integration of Community and Stakeholder Input

Future development could incorporate structured qualitative information from communities and stakeholders.

Potential inputs could include:

- Community safety concerns
- Public survey responses
- Pedestrian and cyclist feedback
- Local accessibility concerns
- Neighborhood priorities

However, this should be implemented carefully. Qualitative inputs should be transparent and governed by a documented methodology rather than being introduced as arbitrary weights that can distort the analytical results.

Final Conclusion

The Vision Zero Chicago project demonstrates how **data analytics, geospatial technology, statistical forecasting, economic evaluation, optimization, and interactive software** can be integrated to support complex public-sector investment decisions.

The project's main achievement is not simply the creation of a predictive model or dashboard. Its stronger contribution is the development of a complete analytical chain:

Data → Risk → Intervention → Benefits → Constraints → Portfolio Decision → Stakeholder Review

The system provides a transparent framework for identifying which potential investments offer the strongest combination of safety impact, economic value, and equity performance under limited capital resources.



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At the same time, the project maintains an important governance boundary:

The system supports decision-making; it does not replace professional engineering judgment or approve projects.

Future development should therefore focus on improving the quality of exposure data, engineering feasibility information, treatment alternatives, and multi-year planning capability.

Ultimately, the Vision Zero Chicago Decision Support System provides a foundation for more **transparent, reproducible, and evidence-based road-safety investment decisions**, while preserving the role of transportation engineers, city officials, and communities in the final decision-making process.



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